

Autoregressive Modeling Across Scales for 3D Generation and Understanding

ABSTRACT

3D structure modeling is essential across scales, enabling applications from fluid simulation and 3D reconstruction to protein folding and molecular docking. Yet, despite shared 3D spatial patterns, current approaches remain fragmented, with models narrowly specialized for specific domains and unable to generalize across tasks or scales. We propose Uni-3DAR, a unified autoregressive framework for cross-scale 3D generation and understanding. At its core is a coarse-to-fine tokenizer based on octree data structures, which compresses diverse 3D structures into compact 1D token sequences. We further propose a two-level subtree compression strategy, which reduces the octree token sequence by up to 8x. To address the challenge of dynamically varying token positions introduced by compression, we introduce a masked next-token prediction strategy that ensures accurate positional modeling, significantly boosting model performance. Extensive experiments across multiple 3D generation and understanding tasks, including small molecules, proteins, polymers, crystals, and macroscopic 3D objects, validate its effectiveness and versatility. Notably, Uni-3DAR surpasses previous state-of-the-art diffusion models by a substantial margin, achieving up to 256% relative improvement while delivering inference speeds up to 21.8x faster.

1 INTRODUCTION

3D structure modeling underpins a wide range of real-world applications, spanning the planetary-scale dynamics of celestial bodies to the angstrom-scale arrangements of atoms and electrons. At the macroscopic level, it enables 3D object reconstruction, computational fluid dynamics simulations, and climate forecasting; at the microscopic level, it supports protein structure prediction (Jumper et al., 2021), crystal generation (Jiao et al., 2023), molecular dynamics (Wang et al., 2018a), and molecular docking (Alcaide et al., 2024).

Despite these shared spatial principles, 3D modeling tasks have largely evolved in silos. Models tailored for macroscopic structures fail to transfer to microscopic domains, and even applications at similar scales rarely generalize. For instance, a model designed for crystal generation cannot be directly applied to protein folding (Xie et al., 2021; Jiao et al., 2023). This fragmented development hinders data reuse and results in redundant, highly specialized models rather than a unified solution.

To overcome this fragmentation, we propose Uni-3DAR, a unified autoregressive framework for cross-scale 3D generation and understanding. At its core is a tokenizer that efficiently compresses diverse 3D structures into discrete 1D token sequences. Leveraging these compressed sequences, our autoregressive model unifies generative and understanding tasks within a single architecture.

The proposed tokenizer uses an octree data structure to compress the full-size 3D grid both losslessly and efficiently. As illustrated in Fig. 2 (a) and (b), we construct an octree by recursively subdividing the space up to a maximum depth of L . To adapt to data sparsity, branches corresponding to empty regions are pruned, resulting in a maximum of 8^{L-1} leaf grid cells (but most will be pruned due to sparsity). We then introduce a fine-grained tokenization that encodes details within each occupied leaf cell (we call it a “3D patch”), such as atomic types and precise coordinates for molecules, or more general VQVAE tokens (Van Den Oord et al., 2017). Concatenating these tokens level by level produces a hierarchical, coarse-to-fine 1D token sequence that effectively represents the 3D structure (fig. 2(c)). Furthermore, we compress each two-level subtree (eight subcells) into a single 8-bit token instead of assigning an individual occupancy token to each node (Fig. 2 (d)). Since each

[Figure 1: Uni-3DAR Overview. (a) A diagram showing the Uni-3DAR architecture. On the left, '3D Structures from Micro to Macro' (a protein, a molecule, a cat) are processed through 'Coarse-to-fine subdivision' and a 'Tokenizer' (Octree + 2-level subtree compression + fine-grained leaf tokens) to produce 'Compressed Spatial Tokens'. These are fed into a 'Model' (Autoregressive Transformer + Masked Next-Token Prediction). The model can be conditioned on 'Cross-Modal Inputs' (Natural language, Biology sequences, Spectral signal) via an 'Optional' path. The output is a 'Unified Framework for Various Tasks' including Single-Frame Generation, Multi-Frame Generation, Token-Level

Understanding, and Structure-Level Understanding. (b) A diagram showing the hierarchy of octree levels: Level 5 (1x1x1), Level 4 (2x2x2), Level 3 (4x4x4), Level 2 (8x8x8), Level 1 (16x16x16), and Level 0 (fingerprint atoms).] (9ba3dc91984c80b96f217fb1bddd5c06_img.jpg)

Figure 1: Uni-3DAR Overview. (a) A diagram showing the Uni-3DAR architecture. On the left, '3D Structures from Micro to Macro' (a protein, a molecule, a cat) are processed through 'Coarse-to-fine subdivision' and a 'Tokenizer' (Octree + 2-level subtree compression + fine-grained leaf tokens) to produce 'Compressed Spatial Tokens'. These are fed into a 'Model' (Autoregressive Transformer + Masked Next-Token Prediction). The model can be conditioned on 'Cross-Modal Inputs' (Natural language, Biology sequences, Spectral signal) via an 'Optional' path. The output is a 'Unified Framework for Various Tasks' including Single-Frame Generation, Multi-Frame Generation, Token-Level Understanding, and Structure-Level Understanding. (b) A diagram showing the hierarchy of octree levels: Level 5 (1x1x1), Level 4 (2x2x2), Level 3 (4x4x4), Level 2 (8x8x8), Level 1 (16x16x16), and Level 0 (fingerprint atoms).

Figure 1: **Uni-3DAR Overview.** (a) A coarse-to-fine octree-based tokenizer converts 3D structures into 1D sequences (details in fig. 2). The tokens are modeled by an autoregressive transformer with masked next-token prediction (details in fig. 3) and can be optionally conditioned on cross-modal inputs (e.g., text, biological sequences, spectra). A single model supports single- and multi-frame generation as well as token- and structure-level understanding. (b) An example of octree from coarse level to fine level. Uni-3DAR generates tokens in a coarse-to-fine order: high-level occupancy tokens followed by level-0 tokens that capture local details (e.g., atom types and coordinates). The merits of octree over other 3D representations are discussed in Appendix A.

subcell is binary (empty or not), grouping eight subcells yields $2^8 = 256$ distinct states, reducing the sequence length approximately 8x and converting 8 binary classifications into one 256-class task.

However, the octree compression with empty tokens pruned disrupts the spatial mapping, meaning adjacent tokens no longer correspond to uniform intervals in the original 3D space. Unlike in 2D images with fixed patch positions, the model cannot reliably predict the next token without knowing its explicit target coordinates. We found that simply appending the next position to the current token yielded unsatisfactory results. To address this challenge, we propose a **masked next-token prediction** strategy. As illustrated in Fig. 3 (a), our method duplicates each token so that it appears twice with the same positional embedding. We then replace the first copy with a [MASK] token. The model still performs next-token prediction, but the prediction is made exclusively at the masked position. This setup ensures that the prediction is conditioned on the correct positional information of the intended target token, effectively resolving the issue of dynamic token positions. Although this approach doubles the sequence length, it achieves significant performance gains as validated in appendix D.2.

Uni-3DAR is built on several technical innovations: (1) a **coarse-to-fine octree-based tokenization** for efficient representation, (2) a **2-level subtree compression** to reduce sequence length, (3) a **unified fine-grained structural representation** (for “3D patch”) to capture local details, and (4) a **masked next-token prediction** strategy to handle dynamic token positions, which enable our key contributions:

- 1. **Unified Cross-Scale 3D Modeling.** Leveraging the proposed coarse-to-fine tokenizer, Uni-3DAR can process a wide range of 3D structures, from macroscopic to microscopic scales.
- 2. **Unified Generation and Understanding.** Uni-3DAR seamlessly unifies 3D structural generation and understanding tasks within a single framework. As illustrated in Fig. 3 (b), different tasks use distinct tokens, ensuring clear separation without interference.
- 3. **High Efficiency.** Thanks to the octree and two-level subtree compression, Uni-3DAR represents 3D structures with far fewer tokens. For example, while O Pinheiro et al. (2023) requires $32^3 = 262,144$ tokens for a small molecule, Uni-3DAR needs only hundreds, and can scale to large proteins with thousands of atoms with deeper octree levels (section 3.4). Moreover, appendix D.3 shows that Uni-3DAR is approximately 21.8x faster than prior diffusion-based models.

! [Diagram (a) shows the coarse-to-fine subdivision of a 2D grid. It starts with a level 0 cell of length 8 containing two atoms (H) at (4, 4) and (6, 6). This is partitioned into four level 1 cells of length 4. The cell containing (4, 4) is further partitioned into four level 2 cells of length 2. The diagram shows the hierarchy of cells and the atoms they contain.

Diagram (b) shows the octree compression process. It starts with the root node (level 0) at (4, 4). This is compressed into a sequence of nodes: (2, 2), (6, 2), (2, 6), (6, 6) at level 1. These are further compressed into level 2 nodes: (1, 1), (3, 1), (1, 3), (3, 3), (5, 5), (7, 5), (5, 7), (7, 7). The (3, 3) node is further compressed into level 3 nodes, which are then converted into fine-grained tokens. A note indicates that in-cell positions are discretized into $S(0.50)^5$ with a default

value of 25. Table showing Octree Spatial Tokens. The columns represent the sequence order of nodes (level-wise). The rows show the position (Pos), level, type (H for atom, empty for cell), in-cell X, and in-cell Y coordinates. Diagram (d) shows 2-level subtree compression. It takes a subtree of nodes (Pos: (4, 4), (2, 2), (6, 2), (2, 6), (6, 6)) and compresses it into a sequence of bits (0001, 1000). These bits are then used to generate compressed tokens.]

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(a) Coarse-to-fine Subdivision

(b) Octree Compression

(c) Octree Spatial Tokens

Pos	(4, 4)	(2, 2)	(6, 2)	(2, 6)	(6, 6)	(1, 1)	(3, 1)	(1, 3)	(3, 3)	(5, 5)	(7, 5)	(5, 7)	(7, 7)	(3, 3)	(5, 5)
Level	0	1	1	1	1	2	2	2	2	2	2	2	2	3	3
Type	1	1	0	0	1	0	0	0	1	1	0	0	0	H	H
In-cell X	25	25	25	25	25	25	25	25	25	25	25	25	25	12	36
In-cell Y	25	25	25	25	25	25	25	25	25	25	25	25	25	25	32

(d) 2-Level Subtree Compression

Diagram (a) shows the coarse-to-fine subdivision of a 2D grid. It starts with a level 0 cell of length 8 containing two atoms (H) at (4, 4) and (6, 6). This is partitioned into four level 1 cells of length 4. The cell containing (4, 4) is further partitioned into four level 2 cells of length 2. The diagram shows the hierarchy of cells and the atoms they contain. Diagram (b) shows the octree compression process. It starts with the root node (level 0) at (4, 4). This is compressed into a sequence of nodes: (2, 2), (6, 2), (2, 6), (6, 6) at level 1. These are further compressed into level 2 nodes: (1, 1), (3, 1), (1, 3), (3, 3), (5, 5), (7, 5), (5, 7), (7, 7). The (3, 3) node is further compressed into level 3 nodes, which are then converted into fine-grained tokens. A note indicates that in-cell positions are discretized into $S(0.50)^5$ with a default value of 25. Table showing Octree Spatial Tokens. The columns represent the sequence order of nodes (level-wise). The rows show the position (Pos), level, type (H for atom, empty for cell), in-cell X, and in-cell Y coordinates. Diagram (d) shows 2-level subtree compression. It takes a subtree of nodes (Pos: (4, 4), (2, 2), (6, 2), (2, 6), (6, 6)) and compresses it into a sequence of bits (0001, 1000). These bits are then used to generate compressed tokens.

Figure 2: Overview of Uni-3DAR tokenization (illustrated in 2D using quadtree for clarity). (a) Adaptive coarse-to-fine subdivision of grid cells, where darker nodes indicate non-empty cells that can be further partitioned. (b) This partitioning process constructs an octree, providing a lossless compression of the full-size 3D grid. (c) Uni-3DAR’s tokenization consists of two components: hierarchical spatial compression via an octree and fine-grained structural tokenization. Each node’s position is determined by its tree level and cell center. (d) The proposed 2-level subtree compression reduces the octree tokens by 8x (4x in the illustrated quadtree).

4. *High Accuracy.* Extensive experiments across diverse tasks—including macroscopic 3D shape generation (table 4), molecular (table 7), crystal generation (table 2), protein pocket prediction (table 6), molecular docking (table 7), and molecular pretraining (tables 8 and 9)—demonstrate Uni-3DAR’s superior or competitive performance compared to existing methods. Notably, Uni-3DAR consistently outperforms diffusion-based models. Ablation studies (table 12) highlight the benefits of unifying generation and understanding and validate the effectiveness of each component.

2 METHOD

2.1 DYNAMIC COARSE-TO-FINE TOKENIZATION FOR 3D STRUCTURES

3D structures are inherently sparse: at the microscopic scale, most space is empty except for scattered atoms; at the macroscopic level, detailed representations are only needed at object surfaces, with most volume remaining empty. Using a full-size voxel grid is thus highly inefficient. To address this, we propose a hierarchical, coarse-to-fine tokenization of 3D structures that exploits this sparsity. As shown in Fig. 1, our approach consists of two parts: (1) a hierarchical compression of 3D space using an octree, and (2) a fine-grained structural tokenization.

The first component is the octree, an efficient data structure for lossless 3D grid compression. Starting with a single cell covering the entire structure, we recursively subdivide it: if a cell contains atoms, it is partitioned further. Each

subdivision halves each dimension, producing $2^3 = 8$ equal subcells (hence “octree”). This process continues for L levels. If c_0 is the root cell length, the

[Figure 3: (a) Masked Next-Token Prediction. (b) Unified Framework for 3D GU.]
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Figure 3 consists of two parts. Part (a) titled "(a) Masked Next-Token Prediction" shows three diagrams. The first, "Next-Token Prediction", shows a sequence of tokens with the current position (Cur Pos: (6,6), (3,3), (5,5)) and the next position (Next Pos: (3,3), (5,5)), noting it "Requires additional next pos". The second, "Masked Prediction", shows a sequence with two masked tokens (m) and the note "Cur pos only, parallel independent sampling". The third, "Masked Next-Token Prediction (Our)", shows a sequence with a masked token (m) and the note "Cur pos only, causal sampling". Part (b) titled "(b) Unified Framework for 3D GU" shows a sequence of tokens. It includes a "condition" (three green squares), "structure 1's octree tokens" (blue squares), "structure 1's fine-grained tokens" (green squares), a "[eos]" token, "structure 2's tokens" (blue squares), and another "[eos]" token. Arrows indicate "Generation (Next-Token Prediction)" (green arrows), "Token-Level Understanding" (green arrows), and "Structure-Level Understanding" (blue arrows). A box labeled "token-level properties" is above the fine-grained tokens, and a box labeled "structure-level properties" is above the structure 2's tokens.

Figure 3: (a) Masked Next-Token Prediction. (b) Unified Framework for 3D GU.

Figure 3: (a) **Masked Next-Token Prediction.** To handle the challenge of dynamically positioned tokens in sparse 3D structures, Uni-3DAR decouples position and content generation. Unlike standard next-token prediction, we first infer the next token’s position from the octree hierarchy, place a “[MASK]” token, and then have the model predict only its content (e.g., occupancy or fine-grained properties). (b) **Unified Framework for 3D Generation and Understanding.** The Uni-3DAR architecture is a versatile, multi-task model. It supports autoregressive generation of complex 3D structures (blue arrows) and can be prompted to perform both token-level (green arrows) and structure-level (blue box) understanding tasks within a single framework.

cell length at level $L - 1$ is $c_{L-1} = c_0/2^{L-1}$. We refer to these leaf subcells as fine-grained “3D patches,” which are then tokenized as detailed in the following paragraph.

The second component is the fine-grained tokenization of structural details. While the octree effectively identifies coarse, non-empty regions, it lacks finer details such as atom types and precise coordinates (microscopic) or surface features (macroscopic). Although using deeper octrees can capture more detail (Ibing et al., 2023), this approach becomes inefficient due to the rapidly increasing number of tokens. Instead, inspired by the use of 2D image patches (Alexey, 2020), we treat the contents of each final-level non-empty region as a “3D patch.” These patches can be processed in various ways; for instance, they can be quantized into discrete tokens for autoregressive prediction, similar to VQ-VAE (Van Den Oord et al., 2017), or modeled using a patch-level diffusion loss for continuous vector representations (Li et al., 2024b) (ablation in Table 12). In our experiments, we demonstrate this flexibility by using raw atom types and coordinates as fine-grained tokens for microscopic data (we set the patch size to ensure each 3D patch only contains one atom), and VQ-VAE tokens for macroscopic data. More details are in Appendix B.

Finally, we concatenate tokens level by level. Beyond token content, we represent each token’s positional information using its tree level and the spatial coordinates of its cell center. For instance, the root cell is at level 0 with a center at $(c_0/2, c_0/2, c_0/2)$. During autoregressive prediction, since octree tokens are dynamically unfolded level by level, the positions of all tokens at the current level are known based on the predictions from previous level. This explicit knowledge of the token position is crucial, as autoregressive models predict only token content.

2-Level Subtree Compression Although octree tokenization avoids cubic cell growth, it remains inefficient for large 3D structures. Each level has up to 8^L tokens (N = non-empty final-level cells), totaling up to 8^{NL} tokens across L levels, about two orders of magnitude larger than N^L . To reduce this, we introduce 2-level subtree compression, merging a parent and its 8 children into a single token. As the parent’s type is always I, the subtree is fully represented by its 8 children’s types, yielding $2^8 = 256$ possible states. This cuts token count by $8 \times$, down

to at most $\mathcal{N}(L - 1)$ tokens. For positional information of the compressed nodes, we retain the their parent’s center and level.

2.2 MASKED NEXT-TOKEN PREDICTION FOR DYNAMIC TOKEN POSITIONS

In standard autoregressive models, such as those for text, token positions follow a fixed, sequential order (e.g., token $S_i + 1$ always follows token S_i). This static structure makes the next token’s position implicitly known, obviating the need for its explicit prediction. In contrast, our coarse-to-fine 3D tokenization generates a token sequence where positions are *dynamic* and vary across different structures. This variability introduces a significant challenge: inferring the next token’s position becomes non-trivial, making it preferable to provide this information to the model explicitly.

A straightforward approach is to encode both the current and next positions within each token (Ibing et al., 2023). However, we found this method leads to suboptimal performance (Table 12). We hypothesize that the unpredictable nature of the next token’s position introduces noise that degrades the current token’s representation. This intuition is supported by prior work like Yan et al. (2022), which decouples position and content prediction into two separate transformer modules.

Another promising direction is masked prediction (Chang et al., 2022; Li et al., 2024b), where a model predicts the content of a masked token given its position. This has proven effective for generative tasks with non-sequential or randomized token orders (Li et al., 2024b; Pang et al., 2024). However, directly applying conventional masked prediction to our framework is problematic. First, it typically relies on bi-directional attention, whereas our hierarchical tokenization unfolds uni-directionally. Second, it often requires parallel, non-causal sampling, which necessitates complex, rule-based inference strategies to balance performance and efficiency (Li et al., 2024b).

To resolve these issues, we introduce **Masked Next-Token Prediction (MNTP)**, a simple yet effective method that integrates masked prediction into a standard autoregressive framework. The core idea is to duplicate each token. For a given token, we first generate a placeholder with its content replaced by a special [MASK] symbol while retaining its correct position. This is immediately followed by a second token at the *same position* but with the true content. The model’s objective is to predict the content of this second token, conditioned on the [MASK] token and all preceding tokens.

This formulation effectively reframes next-token prediction as a masked prediction task: the model is given a position with a mask and asked to fill in the content. This approach elegantly sidesteps the challenge of predicting dynamic next positions. Compared to conventional masked prediction, MNTP preserves a strictly causal, uni-directional attention flow, eliminating the need for complex sampling schemes. While this duplication doubles the sequence length, we demonstrate in Appendix D.2 that the substantial performance gains justify this trade-off. Furthermore, through targeted optimizations discussed in Appendix D.3, the impact on inference latency is modest, with only a 15%–30% increase compared to standard next-token prediction (appendix D.3).

2.3 UNIFIED 3D GENERATION AND UNDERSTANDING FRAMEWORK

By integrating techniques in sections 2.1 and 2.2, Uni-3DAR provides a unified framework for a wide range of 3D tasks (Figure 3(b)). The model architecture assigns distinct roles to different token types, enabling it to handle four primary task categories individually or jointly:

1. **Single-Frame Generation** (sections 3.1, 3.3, 3.4 and 3.6): Generating a single 3D structure, either unconditionally or conditioned on external modalities like text or chemical properties. This is accomplished using the masked tokens for autoregressive generation.
2. **Multi-Frame Generation** (sections 3.2 and 3.5) Autoregressively producing a sequence of 3D structures (multiple otrees), such as a molecular dynamics trajectory, molecular docking, or pocket-based generation. Each frame is distinguished by a unique frame-index embedding.
3. **Token-Level Understanding** (section 3.4) Predicting properties of local components (e.g., atomic forces or partial

4. ****Structure-Level Understanding**** (section 3.6): Predicting global properties of an entire structure (e.g., solubility, toxicity) via a prediction head on the final "[EOS]" token. This allows Uni-3DAR to be pre-trained on large-scale unlabeled 3D data and efficiently fine-tuned for downstream tasks.

Furthermore, the autoregressive nature of Uni-3DAR inherently supports multi-modal conditioning, which is critical for many scientific applications. For instance, a protein’s amino acid sequence can guide the generation of its 3D fold. Similarly, experimental data like Powder X-ray Diffraction (PXRD) spectra can constrain the prediction of a crystal structure, a task we explore in section 3.2.

We conducted extensive experiments to validate Uni-3DAR across diverse benchmarks. This section summarizes the key findings; complete implementation details settings can be found in appendices B and C. To ensure a fair comparison against existing methods, we trained separate model for each

Data	QM9				DRUG		
- - - - - - -							
Atom Sta(%) $\uparrow$ Mol Sta(%) $\uparrow$ Valid(%) $\uparrow$ $\$V \times U(\%) \uparrow$ Atom Sta(%) $\uparrow$ Valid(%) $\uparrow$							
99.0 95.2 97.7 97.7 86.5 99.9							
ENF (Garcia Satorras et al., 2021) 85.0 4.9 40.2 39.4 - -							
G-SchNet (Gebauer et al., 2022) 95.7 68.1 85.5 80.3 - -							
GDM (Hoogetboom et al., 2022) 97.0 63.2 - - 75.0 90.8							
GDM-AUG (Hoogetboom et al., 2022) 97.6 71.6 90.4 89.5 77.7 91.8							
EDM (Hoogetboom et al., 2022) 98.7 82.0 91.9 90.7 81.3 92.6							
EDM-Bridge (Wu et al., 2022) 98.8 84.6 92.0 90.7 82.4 92.8							
GeoLDM (Xu et al., 2023b) 98.9 89.4 93.8 92.7 84.4 99.3							
UniGEM* (Feng et al., 2024) 99.0 89.8 95.0 93.2 85.1 98.4							
Uni-3DR 99.4 93.7 98.0 94.0 85.5 99.4							

Data	Method	Validity (%) $\uparrow$	Coverage (%) $\uparrow$	Property $\downarrow$		
Carbon-24	FTCP (Ren et al., 2021)	0.08	0.00	0.00	5.206	19.05
G-SchNet	(Gebauer et al., 2019)	99.94	0.00	0.00	0.9427	1.320
P-G-SchNet	(Gebauer et al., 2019)	48.39	0.00	0.00	1.533	134.7
CDVAE	(Xie et al., 2021)	100.0	99.80	83.08	0.1407	0.2850
DiffCSP	(Jiao et al., 2023)	100.0	99.90	97.27	0.0805	0.0820
Uni-3DR		99.99	100.0	98.16	0.0660	0.0289
MP-20	FTCP (Ren et al., 2021)	1.55	48.37	4.72	0.09	23.71
G-SchNet	(Gebauer et al., 2019)	99.65	75.96	38.33	99.57	3.034
P-G-SchNet	(Gebauer et al., 2019)	77.51	76.40	41.93	99.74	4.04
CDVAE	(Xie et al., 2021)	100.0	86.70	99.15	99.49	0.6875
					0.2778	1.432

et al., 2021; Jiao et al., 2023; Miller et al., 2024), we use Carbon-24 (Pickard, 2020), MP-20 (Jain et al., 2013), and MPTS-52 datasets. De novo generation is evaluated via validity, coverage, and property statistics (Xie et al., 2021), while CSP and PXRD-guided CSP are assessed by top-1 match rate and RMSE, using StructureMatcher (Ong et al., 2013) with the same thresholds as in (Miller et al., 2024).

Table 2 shows Uni-3DAR’s performance on Carbon-24 and MP-20. On Carbon-24, Uni-3DAR outperforms baselines, especially in coverage, generating diverse and realistic structures. On MP-20, it achieves higher component validity while maintaining competitive results overall, highlighting its strength in producing chemically valid crystals. Table 3 summarizes CSP results across all datasets. Uni-3DAR consistently outperforms baselines, improving match rate by 4.14% on Carbon-24 and reducing RMSE from 0.0566 to 0.0317 on MP-20 (178% relative gain). On MPTS-52, it achieves 0.0684 RMSE, a 184% improvement despite higher complexity, demonstrating strong precision and generalization. For PXRD-guided CSP, Uni-3DAR surpasses PXRDGEN (Li et al., 2024a), raising the match rate from 68.68% to 75.08% and cutting RMSE from 0.0707 to 0.0276 (256% relative gain), showing exceptional accuracy in reconstructing crystals from PXRD data.

3.3 MACROSCOPIC 3D OBJECT GENERATION

To demonstrate its versatility beyond microscopic domains, Uni-3DAR was also evaluated on unconditional macroscopic 3D object generation, a fundamental task in 3D computer vision. We utilized three ShapeNet categories (*airplane*, *chair*, *car*) (Chang et al., 2015), where objects are represented as point clouds, and assessed using 1-NNA (with both Chamfer distance (CD) and earth mover distance (EMD) as our main metric following Vahdat et al. (2022)). A distinctive aspect for this

Table 4: Unconditional 3D object generation results (1-NNA) on ShapeNet. The best and second-best results among the baselines are highlighted.

Method	Airplane	Chair	Car
CD ↓	EMD ↓	CD ↓	EMD ↓
r-GAN (Achlioptas et al., 2018)	98.40	96.79	83.69
l-GAN (CD) (Achlioptas et al., 2018)	87.30	93.95	68.58
l-GAN (EMD) (Achlioptas et al., 2018)	89.49	76.91	71.90
PointFlow (Yang et al., 2019)	75.68	70.74	62.84
SoftFlow (Kim et al., 2020)	76.05	65.80	59.21
SerVae (Kim et al., 2021)	76.54	67.65	58.84
DiffNet (Kiston et al., 2020)	75.18	65.55	62.00
DPM (Luo and Hu, 2021)	76.42	86.91	60.05
PVD (Zhou et al., 2021)	73.82	64.81	56.26
LION (Sohoni et al., 2022)	67.41	61.35	55.70
Uni-3DAR (Ours)	67.35	61.09	53.11
	50.89	51.78	53.35

task is Uni-3DAR’s processing of an input object as $512 \times 512 \times 512$ voxels, and the resulting 3D patches (fine-grained structural tokens) are defined as $16 \times 16 \times 16$ voxels. Each patch is quantized using VQVAE. As shown in Table 4, Uni-3DAR exhibits highly competitive, often superior, performance against established baselines (Yang et al., 2019). More details are in Appendix B.

3.4 PROTEIN POCKET PREDICTION

Predicting protein binding pockets is crucial for drug design and molecular docking. We evaluate Uni-3DAR’s token-level understanding on this task, formulating it as a classical atom-level classification problem where each atom is labeled as part of a pocket or not. Following previous work (Zhao et al., 2024), we train and evaluate on a composite dataset built from the CASF-2016 core set (Su et al., 2018), the PDBBind v2020 refined set (pdb, 2025), and MOAD

(Hu et al., 2005). Performance is measured using the Intersection-over-Union (IoU) metric. As shown in Table 6, Uni-3DAR achieves state-of-the-art performance. Notably, it matches or exceeds specialized methods like Vabs-Net, which relies on additional features such as ESM embeddings and Solvent Accessible Surface Area, whereas Uni-3DAR uses only 3D structural information. These results highlight Uni-3DAR’s strong capacity to interpret protein structures for fine-grained, atom-level prediction tasks.

3.5 MOLECULAR DOCKING

Molecular docking, which predicts the binding pose of a ligand to a protein, is a cornerstone of drug discovery. Uni-3DAR frames this as a three-frame generation task: the first two frames are the protein and the initial ligand conformation, and the third is the predicted docked pose. We evaluate this approach on the PDBbind2020 dataset (pdb, 2025), benchmarking against 13 classical and deep learning methods using standard RMSD-based metrics (Top-1/5 success rates for RMSD < 1 Å and < 2 Å, and median RMSD), following the protocol of Cao et al. (2024). Uni-3DAR operates solely on atom types and coordinates, forgoing complex feature engineering and a separate scoring model; poses are ranked using the cumulative probability from the autoregressive generation. The results in Table 7 demonstrate that Uni-3DAR achieves state-of-the-art performance. It surpasses the previous best, SurfDock, on Top-1 metrics, with higher success rates for poses with RMSD < 1 Å (44.75% vs. 40.96%) and < 2 Å (69.06% vs. 68.41%), and a lower median RMSD (1.08 Å vs. 1.18 Å). While its Top-5 performance is slightly lower, likely due to its implicit scoring mechanism, these results underscore the strong potential of our unified, feature-light approach for molecular docking.

3.6 MOLECULAR AND POLYMER PROPERTY PREDICTION VIA PRETRAINING

To evaluate its structure-level understanding, we assess Uni-3DAR on property prediction for small molecules and homopolymers after pretraining. For small molecules, we adopt the pretraining data, downstream tasks, and evaluation settings from state-of-the-art models Uni-Mol (Zhou et al., 2023b) and SpaceFormer (Lu et al., 2025), using Mean Absolute Error (MAE) as the metric. For homopolymers, we follow Wang et al. (2024) and use eight DFT-calculated property datasets, reporting the Root Mean Squared Error (RMSE) from a 5-fold cross-validation averaged over three seeds.

As summarized in Tables 8 and 9, Uni-3DAR demonstrates strong and versatile performance. On small molecule tasks (Table 8), it ranks first in 4 of 10 tasks and in the top two for 8 of 10, performing comparably to the specialized SpaceFormer model. On homopolymer tasks (Table 9), it ranks first in 4 of 8 tasks and in the top two for 7 of 8. These results affirm that Uni-3DAR develops robust and competitive representations for predicting properties across diverse chemical systems.

3.7 ADDITIONAL EXPERIMENTS

Due to space limitations, we present further experimental results in Appendix D. These include (1) an analysis of the benefits of unifying understanding and generation, (2) comprehensive ablation studies evaluating our proposed tokenization and MNTP, and (3) a comparison of inference speeds.

4 RELATED WORK

****Octree and Hierarchical Autoregressive Models**** The coarse-to-fine hierarchical structure is widely used in 3D vision (Wang et al., 2017; Tatarchenko et al., 2017; Tang et al., 2021; Zhou et al., 2023a; Wang, 2023; Ibing et al., 2023; Zhang et al., 2024b; Ren et al., 2024). Among these works, (Ibing et al., 2023) is most similar to Uni-3DAR, as it also employs autoregressive generation using an octree. However, our method differs in three key aspects: (1) instead of relying on deep tree-level generation for fine details, we add an extra layer of fine-grained tokens to avoid excessively deep trees; (2) rather than compressing nodes via convolutional layers, we represent a compressed subtree with a single token; and (3) to handle dynamic token positions, while (Ibing et al., 2023) appends the next position to the current token, we adopt a masked next-token prediction strategy. These innovations make Uni-3DAR more efficient and effective than (Ibing et al., 2023). Recently, some image generative models have adopted a coarse-to-fine, level-by-level

generation approach, such as VAR (Tian et al., 2024). Although the high-level idea appears similar, our motivation is distinct: Uni-3DAR is designed to avoid the inefficiencies of a full-size cubic grid, whereas VAR uses more tokens to boost performance. Moreover, Uni-3DAR remains within the next-token prediction framework, while VAR employs next-scale prediction.

****Microscopic 3D Structure Modeling**** Most previous generative models for microscopic 3D structures employ diffusion-based approaches (Wu et al., 2022; Anand and Achim, 2022; Hoogeboom et al., 2022; Xu et al., 2023b; Jiao et al., 2023) to generate atomic positions from noise. However, diffusion models have two major limitations. First, they require the number of atoms to be predetermined. Second, atom types are sampled from a categorical distribution, for which a proper score function is not well defined. Some studies have explored grid-based generation (O Pinheiro et al., 2024), but using a full-size 3D grid is computationally prohibitive. Other works have

investigated autoregressive models for 3D molecules (Luo and Ji, 2022; Luo et al., 2021; Zhang et al., 2025), but these models generate molecules atom by atom, requiring a predefined sequence order. For microscopic 3D structure understanding, prior studies primarily leverage SE(3)-invariant or equivariant models (Schütt et al., 2021; Fuchs et al., 2020). Additionally, unsupervised pretraining is widely used to mitigate the scarcity of labeled data (Stärk et al., 2022a; Cui et al., 2024; Yang et al., 2024; Zaidi et al., 2022; Zhou et al., 2023b). These models typically follow a BERT-style pretraining framework (Devlin, 2018), where some atoms are masked, their 3D positions are perturbed, and the model is trained to recover the ground truth. While highly effective for understanding tasks, most of these models cannot be directly applied to generation.

Some recent efforts have attempted to unify generation and understanding for microscopic data. However, most focus solely on sequence data (e.g., 1D SMILES, nucleotide sequences, or textual descriptions) and directly apply autoregressive language models (Christofidellis et al., 2023; Zhang et al., 2024a; Nguyen et al., 2024; Xia et al., 2025). While these models are straightforward, they lack essential 3D structural information, limiting their performance and applicability. Recent studies have also explored diffusion-based approaches. For example, UniGEM (Feng et al., 2024) demonstrated that a two-phase, multi-task training strategy can improve performance for both tasks. This approach combines diffusion loss with a prediction task applied during later diffusion steps. In summary, while previous work has made progress in bridging generation and understanding, Uni-3DAR is the first autoregressive framework to unify both tasks for 3D microscopic structures.

****Macroscopic 3D Structure Modeling**** Macroscopic 3D structure modeling encompasses the understanding and generation of everyday objects (Chang et al., 2015; Deitke et al., 2023), scenes (Peng et al., 2023), CAD models (Wu et al., 2021; Willis et al., 2021; Xu et al., 2024), avatars (Canfes et al., 2023), and more. Similar to microscopic 3D structures, macroscopic 3D structures lack a unified representation format. Commonly used 3D representations include voxels (Wang et al., 2018b), point clouds (Xue et al., 2023), polygon meshes (Liu et al., 2024), implicit functions (Tang et al., 2021), and 3D Gaussian Splatting (Kerbl et al., 2023). Recent methods (Zhang et al., 2023a; Zhao et al., 2023; Zhang et al., 2024b; Chen et al., 2024) based on Diffusion Transformers (Peebles and Xie, 2023) encode 3D shapes into compressed, compact latent codes, substantially improving representation efficiency. Previous literature also explored autoregressive modeling for macroscopic 3D structures. For example, Polygen (Nash et al., 2020) and MeshGPT (Siddiqui et al., 2024) generate mesh faces sequentially from lowest to highest on the vertical axis, corresponding to the point-based tokenization strategy as discussed in Sec. 1, suffering from the same challenges in dynamic token positions. Another category of 3D structure generation methods, known as optimization-based approaches (Tang et al., 2023; Lin et al., 2023a; Metzler et al., 2023; Poole et al., 2022), leverages text-to-image generative models and refines 3D representations by distilling information from 2D images (Poole et al., 2022). Unlike true 3D generation, these methods primarily perform 3D reconstruction, making them fundamentally distinct from the previously mentioned 3D generation techniques and Uni-3DAR.

5 CONCLUSION

In this work, we introduced Uni-3DAR, a unified autoregressive framework designed to address the long-standing fragmentation of 3D modeling. By leveraging a novel coarse-to-fine octree-based tokenizer, Uni-3DAR compresses diverse 3D structures—from molecules to macroscopic shapes—into a common 1D sequence representation. This core

innovation, enhanced by 2-level subtree compression for efficiency and a masked next-token prediction strategy to handle sparse spatial data, enables a single model to seamlessly bridge the gap between generative and understanding tasks across different scales. Our extensive experiments validate this unified approach, demonstrating that Uni-3DAR achieves state-of-the-art or highly competitive performance on a wide array of benchmarks. Notably, it consistently outperforms specialized, diffusion-based models while being significantly more efficient. Additionally, we prioritize a general, scalable architecture over hard-coded inductive biases like SE(3) invariance. By relying on data augmentation rather than architectural constraints, Uni-3DAR effectively learns geometric principles from data, matching or outperforming specialized SE(3)-equivariant baselines on molecular tasks. By proving that a simple yet powerful autoregressive paradigm can unify disparate tasks without compromising accuracy, we believe Uni-3DAR marks a pivotal step toward a general-purpose foundation model (a “GPT-2 moment”) for the cross-scale 3D domain.

****Limitations**** While our results demonstrate the mutual benefits of unifying generation and understanding, we have not yet trained a single, large-scale foundation model on a heterogeneous mixture of 3D data and tasks. Realizing this vision through joint pretraining is a primary goal for future work. Other critical avenues for research include extending the framework to real-world applications.

ETHICS STATEMENT

The research presented in this paper aims to advance scientific discovery by creating a unified framework for 3D modeling. We have strived to conduct this work with the highest ethical standards.

All datasets used in our experiments—including QM9, GEOM-DRUG, Materials Project, ShapeNet, and PDBBind—are publicly available and are standard benchmarks in their respective scientific communities. We did not collect any new data, and no personally identifiable or sensitive information was used.

We acknowledge that generative models for molecular and material design could potentially be misused for creating harmful substances. However, Uni-3DAR is intended as a fundamental scientific tool to accelerate beneficial research in fields such as drug discovery and materials science. Its capabilities are grounded in the principles learned from public scientific data. We believe the potential benefits—such as the rapid design of novel therapeutics and efficient materials—significantly outweigh the risks. As with any powerful technology, we advocate for its responsible use and encourage the research community to establish clear guidelines for the ethical application of generative models in science.

Finally, we recognize the environmental impact associated with training large-scale models. Our work incorporates significant efficiency optimizations, such as octree and subtree compression, which substantially reduce the computational resources and token count required for training and inference compared to alternative approaches.

REPRODUCIBILITY STATEMENT

To ensure the reproducibility of our results, we are committed to making our research as transparent as possible.

****Code**** Upon acceptance of this paper, we will release the complete source code for Uni-3DAR, including model implementation, training scripts, and evaluation protocols, under a permissive open-source license.

****Data**** All datasets used in our experiments are publicly available and have been cited appropriately. We followed standard data processing and splitting protocols as established in prior work. Detailed descriptions of data preparation for each task are provided in Appendix C.

****Hyperparameters and Architecture**** The full details of our model architecture, as well as the specific hyperparameters used for every experiment (including learning rates, batch sizes, model dimensions, and training steps), are thoroughly documented in Appendix B and C. We used a consistent model configuration across most tasks to demonstrate the robustness and generality of our framework.

****Computational Environment**** Our experiments were conducted using standard deep learning libraries. Specific

details about the hardware (e.g., NVIDIA A100 and 4090 GPUs) and software environment are provided in the appendices to facilitate the replication of our training and inference setups. The efficiency optimizations used, such as FlashAttention and KV-caching, are also described in Appendix B.

REFERENCES

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